Name: **Date:**

Estimation & Measurement Stations – Student Recording Sheet

Station 1 – Mass / Weight

How would you measure the mass/weight of the heavy book and the light book, and which units make the most sense for each one? Would those same units be used for weighing a person?

I WOULD MEASURE THE HEAVY BOOK BY

_____

_____.

I WOULD MEASURE THE LIGHT BOOK BY

_____

_____.

UNITS I USED ARE: HEAVY BOOK: _____ LIGHT BOOK_____

THE UNITS I WOULD USE TO WEIGH PERSON ARE: _____

Station 2 – Length / Height

What tools at your station would you use to measure the height of a classmate and the lemon juice bottle? What units did you measure in and why are those units the best choice?

TOOLS	CLASSMATE	LEMON JUICE BOTTLE	UNITS
Measuring Tape Orange			
Ruler			
Meterstick			
Paperclip			
Pencil			
Measuring Tape Yellow			

THESE UNITS ARE THE BEST CHOICE

_____

_____

Station 3 – Capacity

Estimate how much liquid is in each container at the station. Record this below. Don't forget to include the units you use. **You can measure the water from the bottle to help you.**

ITEM	ESTIMATION OF CAPACITY	UNITS
Canola Oil		
Bottle of Water		
Hand Cream		
Hot Sauce		

How did you know what units to use?

_____

_____

Station 4 – Ingredients (Solids & Liquids)

For each ingredient (solid and liquid), choose the proper measuring tool to get the measurements needed for the recipe below.

Hot Chocolate Christmas Bread

INGREDIENT	MEASURING TOOL USED
5 grams YEAST	
1 cup HOT CHOCOLATE MIX	
2L WATER	
10mL CORN SYRUP	
¼ cup BAKING SODA	

How did you know which tool to use?

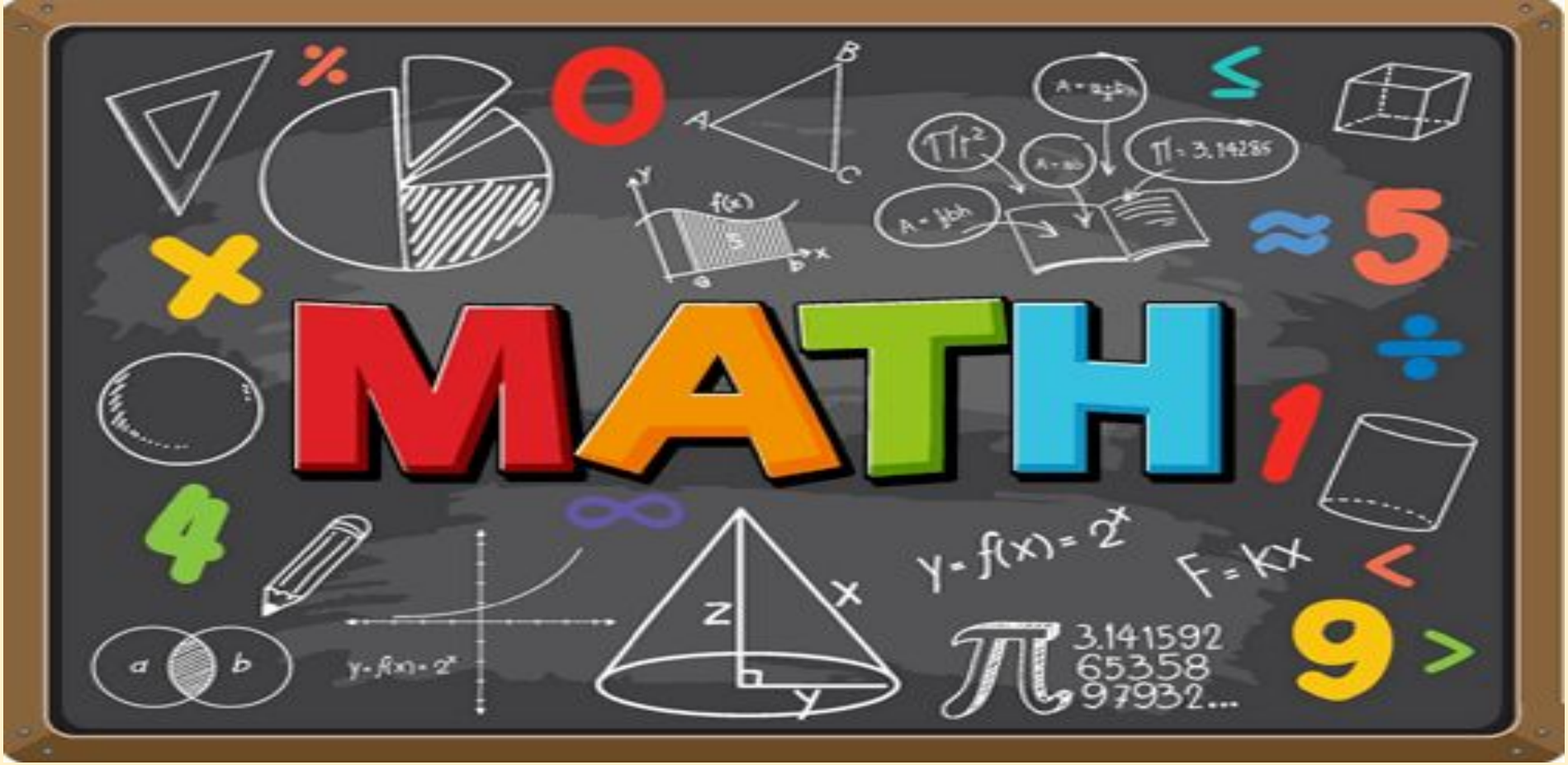
_____

_____

Convert the yeast to mg? *Remember to move the decimal in the right direction.

_____

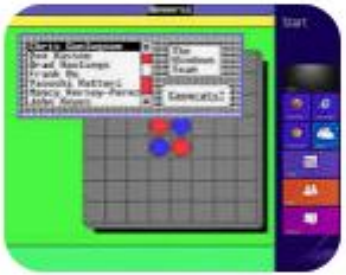
_____



Measurement

On This Day In History

Windows was first released on **November 20, 1985**, with the release of Windows 1.0. This was a graphical operating environment that ran on top of MS-DOS, offering features like a mouse-driven interface and a multitasking graphical user interface.



- **Release Date:** November 20, 1985
- **Version:** Windows 1.0
- **Function:** It was a graphical user interface (GUI) that served as a shell for MS-DOS, allowing users to run graphical programs and use a mouse to navigate.

Learning Goal

- We will be able to understand and accurately use the concepts of mass, capacity, and weight by identifying appropriate units of measurement, selecting suitable tools, choosing units that make sense when estimating, and applying these skills to solve real-life problems.

Review

Concept	Definition	Units
Mass		
Capacity		
Weight		

Review

Question 1

Convert 800 m to _____ km

Question 2

Convert 3 g to _____ mg

Estimation

What comes to mind when you hear the word "Estimation"?

What is Estimation?

- Estimation is a smart, reasonable guess using what we already know
- Estimation is NOT a random guess
- Estimations help us check if future answers make sense before we measure anything

Why Do We Estimate?

- We estimate because it helps us make quick decisions when we do not have the exact measurement tools available or when we do not need an exact answer right away.
- Estimation helps us check whether a final measured answer makes sense or whether it seems too big, too small or unreasonable.
- We use estimation all the time in real life, such as deciding how much you can fit in your desks or whether or not your backpack is too heavy, how much to eat, town speakers

Picking the Right Unit

- Before we estimate anything, we must first pick which unit of measurement makes the most sense for the object we are looking at.
- We always try to match the unit to the size of the object, because using a unit that is too big or small can make the measurement confusing.

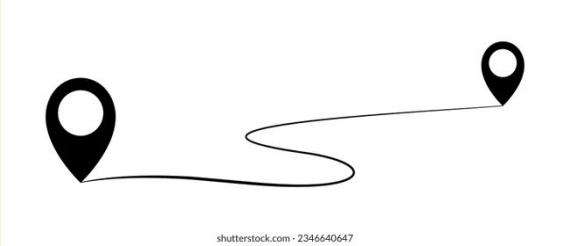
How Do We Make Good Estimates?

- When we try to choose the best unit for something, it helps to think about other objects we already know the size of.
- We look at the new object and ask ourselves, "What does this remind me of?"
- Comparing the new object to something familiar helps us choose the unit that makes the most sense.

Look at these photos, have I estimated correctly?

Mass of feather in cm?

How much is in my yeti in g?



Distance in mL?

What unit should we pick?

Activity

- Get into your designated groups
- As a group you are going to work together for estimating different items, then using different tools to measure or weigh
- At the end your group is going to present to the class your results. Make sure you explain everything!!
- With your group go to your assigned station

Math Centres

12:00

Begin cleaning up when timer turns RED!

Weight	Height	Capacity	Estimation
Centre 1	Centre 2	Centre 3	Centre 5
Student	Student	Student	Student
Student	Student	Student	Student
Student	Student	Student	Student
Student	Student	Student	Student
Student	Student	Student	Student